MODULE 13, LESSON 1

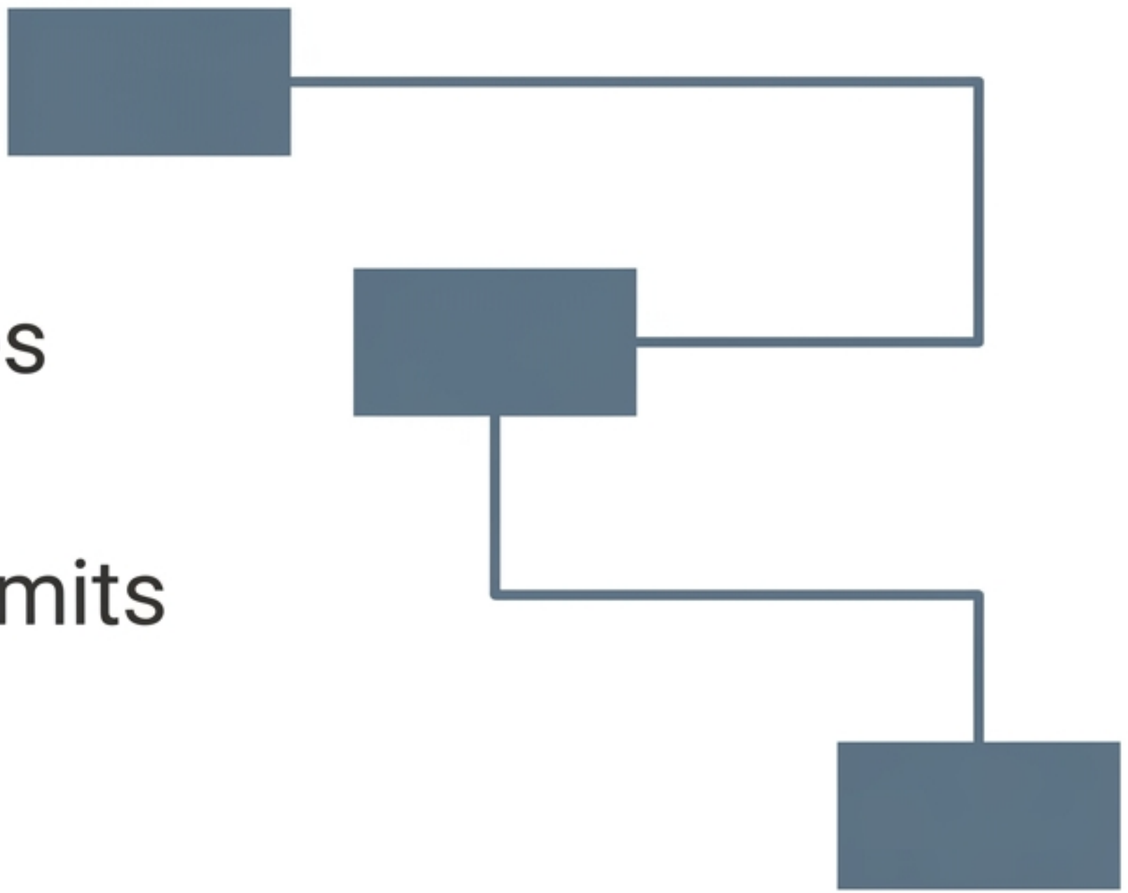
AI 101 for Mechanical Engineers

What a language model produces

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Briefing Structure

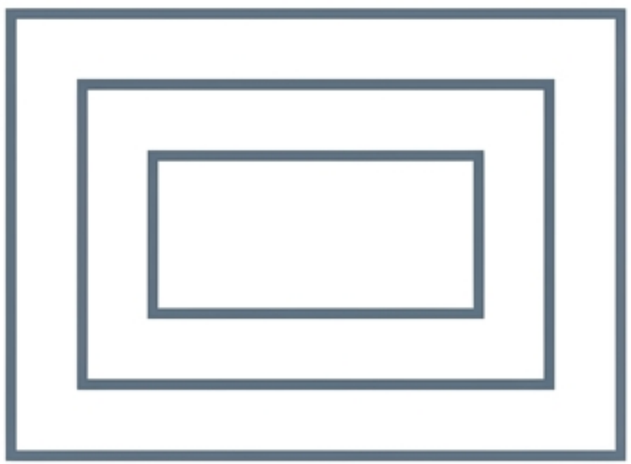
1. System Taxonomies
2. Generative Architectures
3. Operational Evidence Limits



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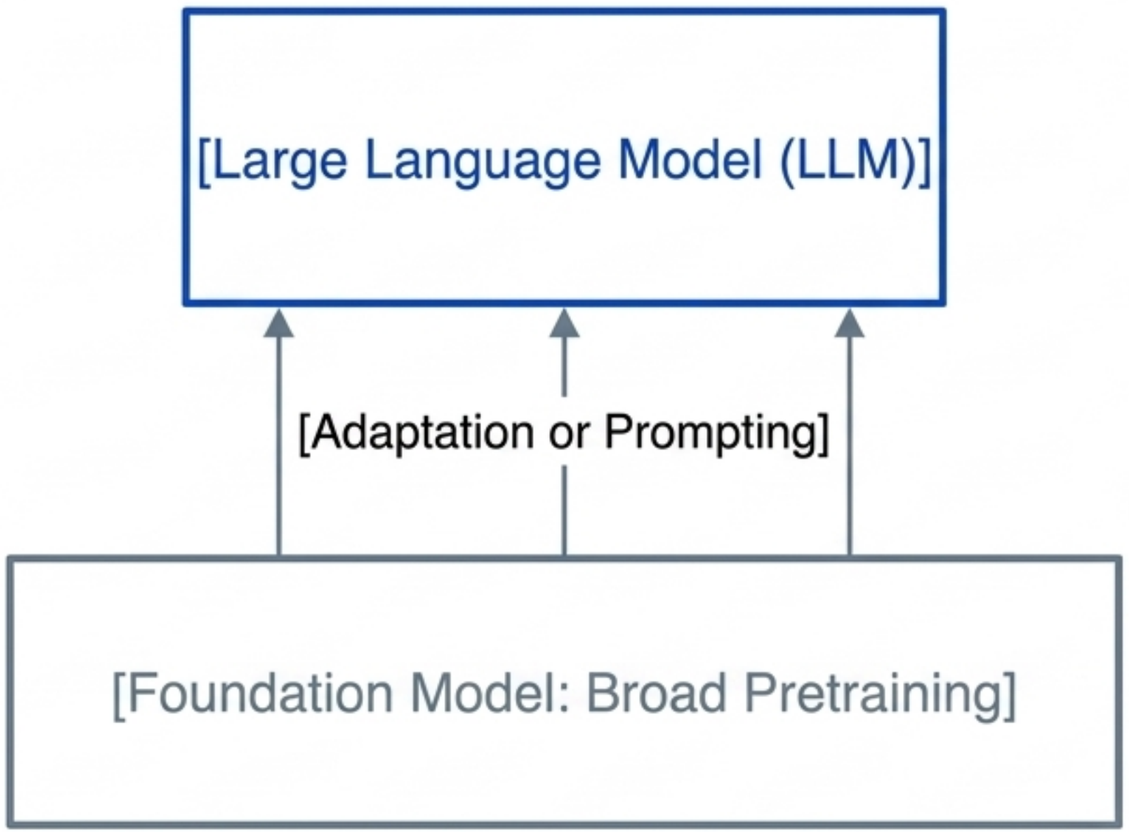
01. System Taxonomies

Establishing the foundational classification of language and multimodal processing systems.



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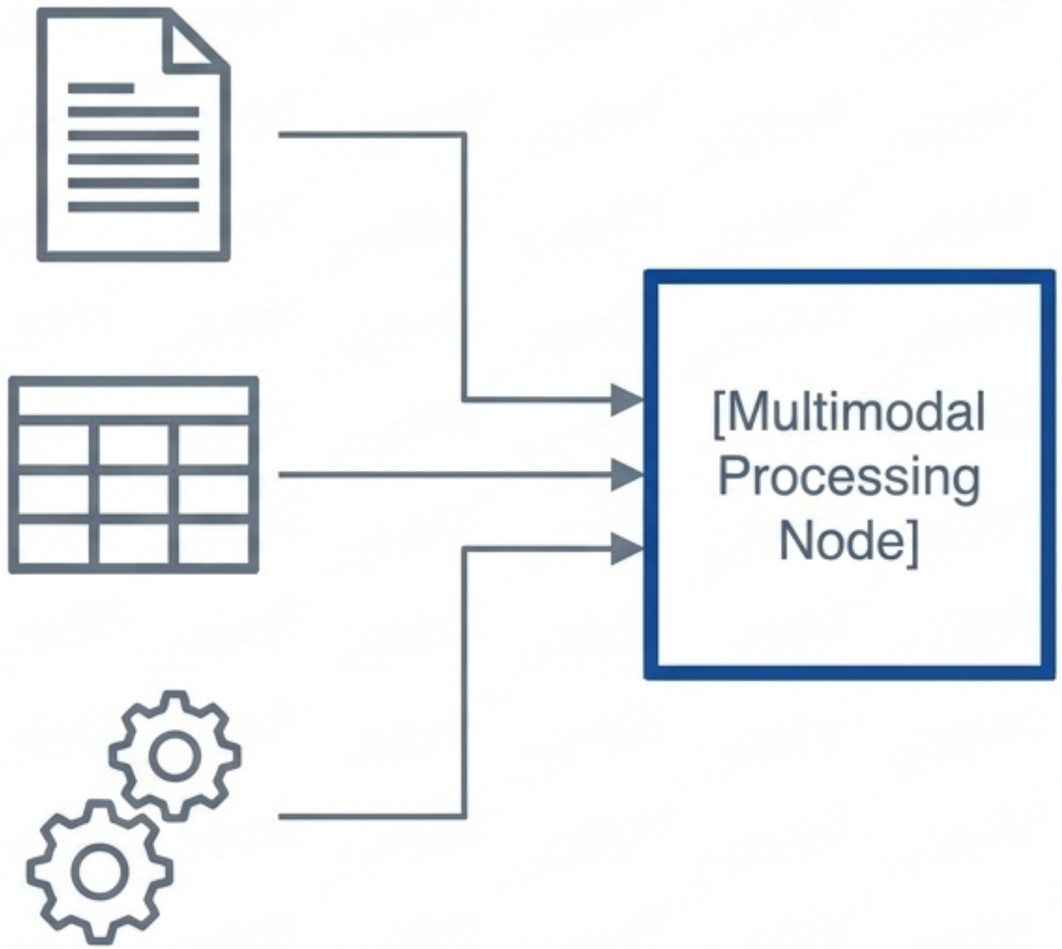
Foundation and Language Systems



- Foundation models are pretrained on broad data for downstream reuse.
- Language models learn patterns over token sequences.
- LLMs process and generate text using high parameter counts.
- Foundation capabilities allow adaptation to specific engineering tasks.

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Multimodal Processing Matrices

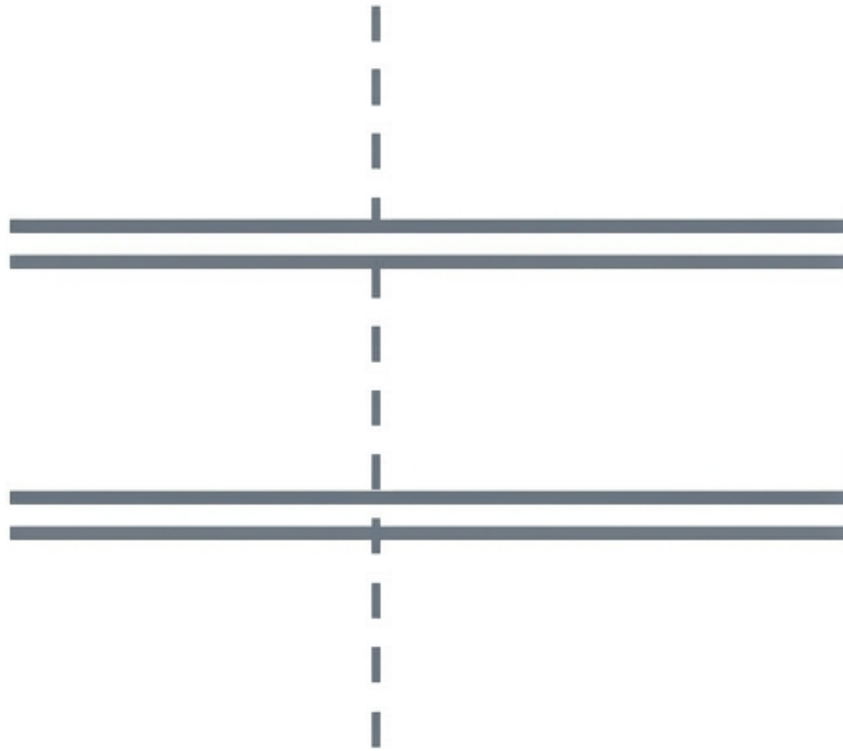


- Multimodal systems process multiple distinct information types simultaneously.
- Input modalities can include both text sequences and image data.
- These models can generate multi-format outputs from mixed inputs.
- Model names describe capabilities, not guarantees of technical correctness.

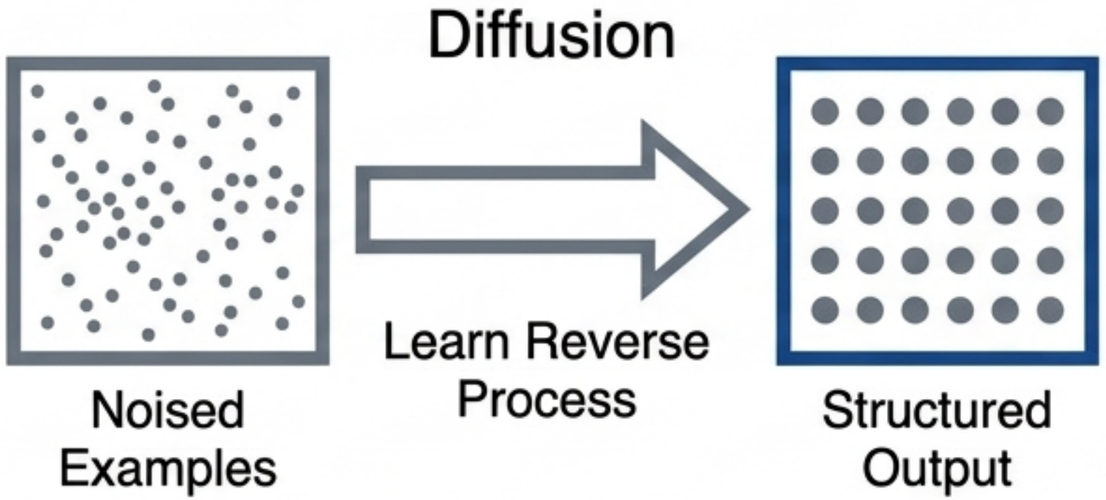
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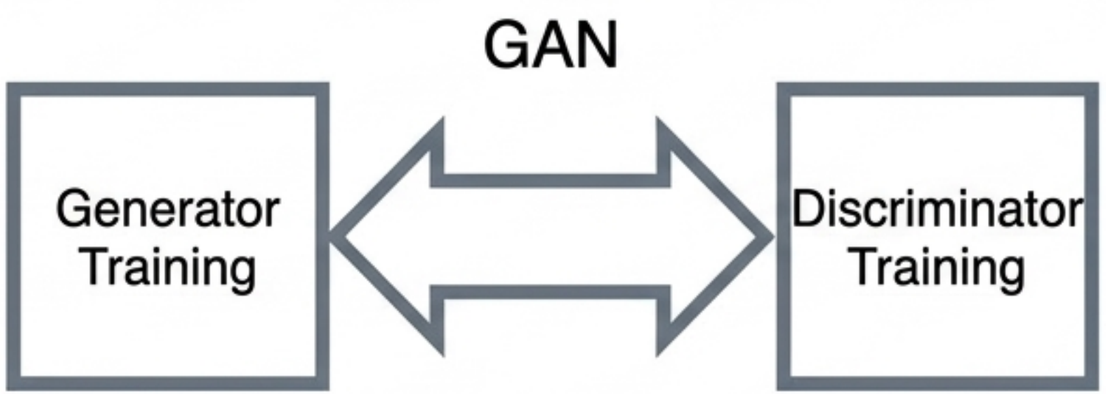
02. Generative Architectures

Outlining the mechanistic approaches used by diffusion systems and generative adversarial networks.



Diffusion and Adversarial Networks





- Diffusion models learn a reverse process from noised data.
- Diffusion architectures are typically used to generate illustrative images.
- GANs utilize concurrent generator and discriminator training phases.
- Detailed mathematical derivations remain outside current scope.

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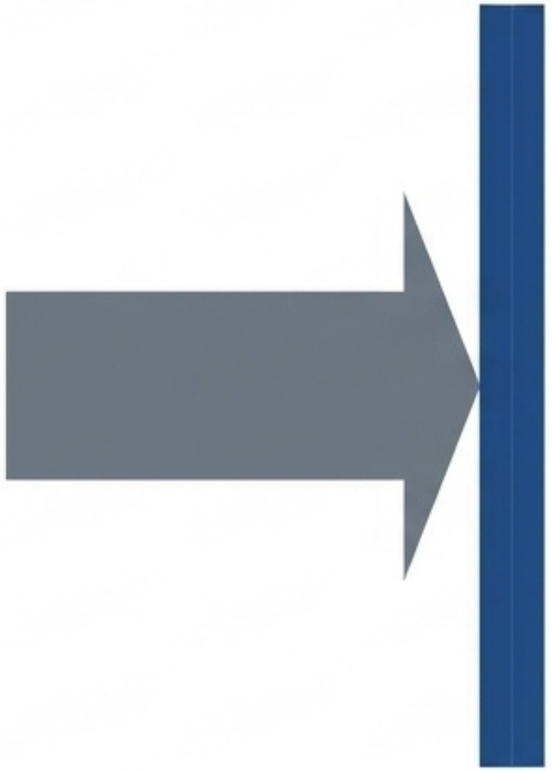
Model Taxonomy Reference Matrix

Model Architecture	Primary Function	Engineering Evidence Limit
Foundation Model	Broad pretraining and adaptation	Task-specific evaluation required
Large Language Model	Processes token sequences	Fluency does not verify claims
Multimodal Model	Handles multiple information types	Capabilities vary by task
Generative Systems	Produces source text or images	Execution and tests required

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03. Operational Evidence Limits

Defining the critical boundaries between generated fluency and **verified engineering evidence**.



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The Evidence Limit Principle

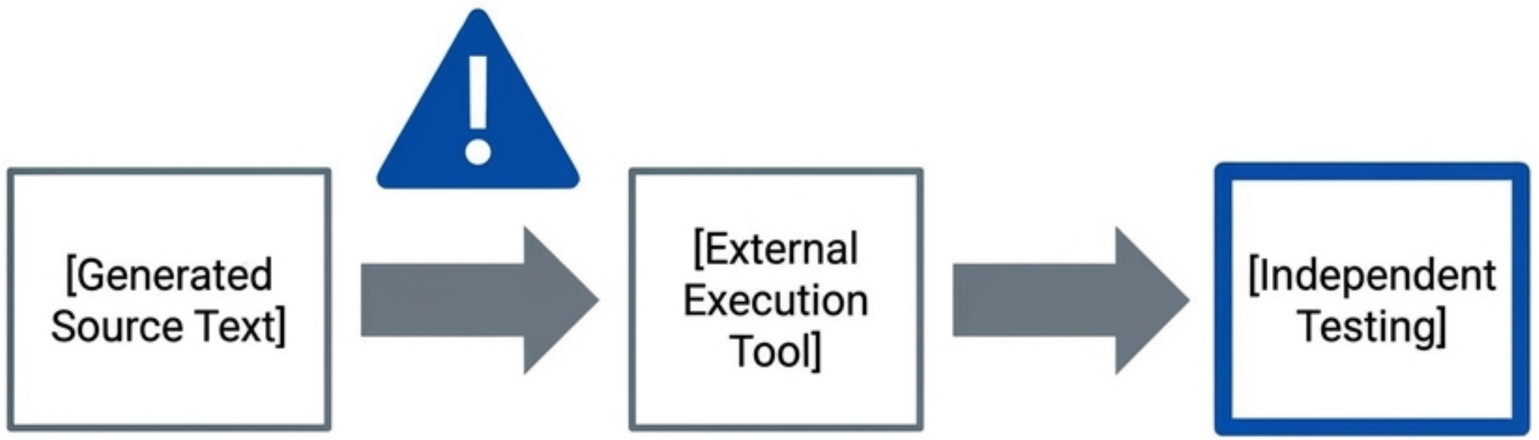




- Generating a plausible continuation is not factual verification.
- Fluency does not validate claims or hardware evidence.
- Generated text cannot substitute consulting an engineering standard.
- Synthetic evidence does not validate real equipment.

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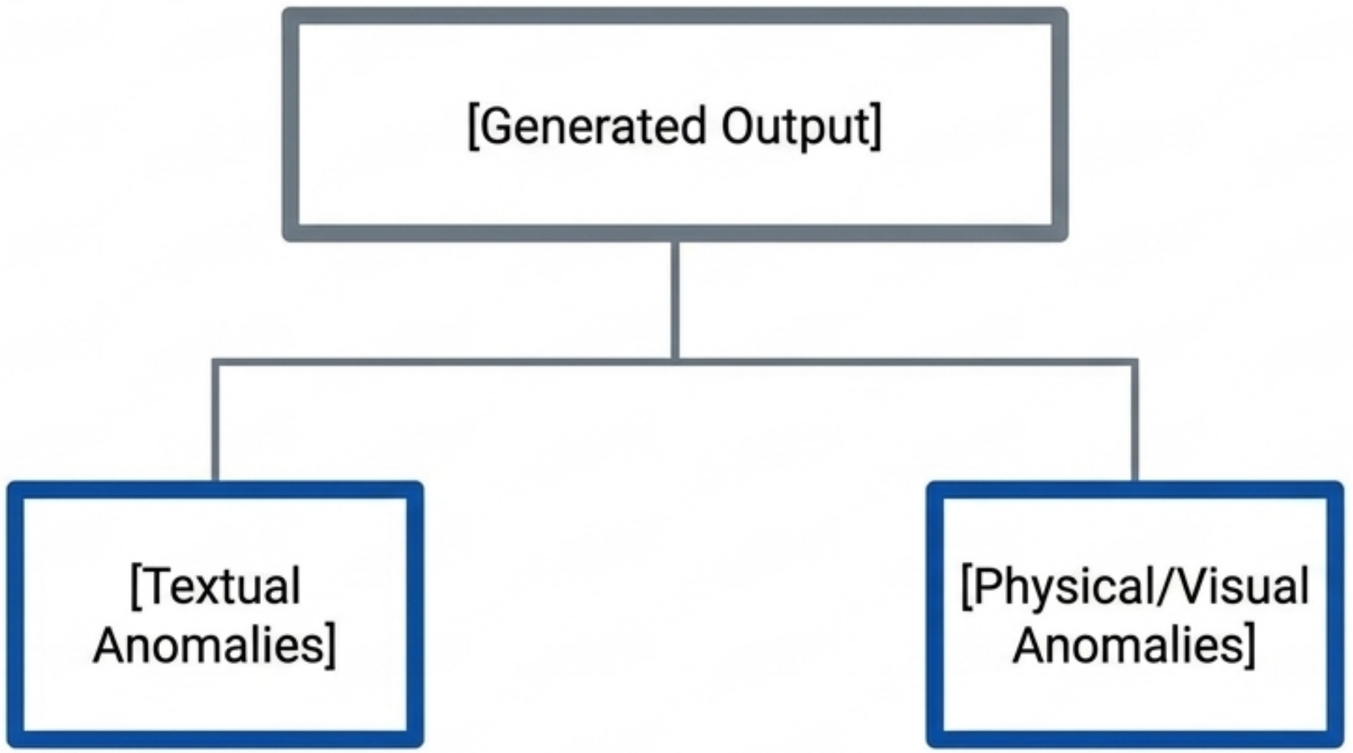
Limits of Code Generation



- Generating code merely creates a sequence of text.
- Text generation does not mean successful execution has occurred.
- An external software tool must execute the generated source.
- Independent tests must verify assumptions and numerical results.

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Factual and Dimensional Constraints



- Fluent explanations may contain entirely incorrect mathematical equations.
- Text generation systems can invent non-existent reference sources.
- Generated analysis may omit critical engineering assumptions.
- Concept drawings frequently fail dimensional, manufacturing, or safety requirements.

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Key Points Summary

1. Taxonomies: Models are classified by training breadth and information modalities.

2. Architectures: Generative systems utilize sequential tokens, diffusion, or adversarial networks.

3. Evidence Limits: Generated text or drawings require rigorous, independent engineering verification.

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